

● MEDIUM

● ALGEBRA

Algebra

10 questions · ~15 min

16

Q1

ALGEBRA

In the xy -plane, line s passes through the point $0,0$ and is parallel to the line represented by the equation $y = 18x + 2$. If line s also passes through the point $4,d$, what is the value of d ?

- A. 2
- B. 18
- C. 72
- D. 74

$$c(x) = mx + 500$$

A company's total cost $c(x)$, in dollars, to produce x shirts is given by the function above, where m is a constant and $x > 0$. The total cost to produce 100 shirts is \$800. What is the total cost, in dollars, to produce 1000 shirts? (Disregard the \$ sign when gridding your answer.)

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

ALGEBRA

$$6x + 7y = 28$$

$$2x + 2y = 10$$

The solution to the given system of equations is x, y . What is the value of y ?

A. -2

B. 7

C. 14

D. 18

Q4

ALGEBRA

In the xy -plane, line t passes through the points 0, 9 and 1, 17. Which equation defines line t ?

A. $y = \frac{1}{8}x + 9$

B. $y = x + \frac{1}{8}$

C. $y = x + 8$

D. $y = 8x + 9$

x	gx
1	54
2	51
3	48
4	45

For the linear function g , the table shows four values of x and their corresponding values of gx . The function can be written as $gx = mx + b$, where m and b are constants. What is the value of b ?

- A. 3
- B. 27
- C. 54
- D. 57

Q6

GRID-IN

ALGEBRA

$$3x + 6 = 4y$$

$$3x + 4 = 2y$$

The solution to the given system of equations is x, y . What is the value of y ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

If the graph of $27x + 33y = 297$ is shifted down 5 units in the xy -plane, what is the y -intercept of the resulting graph?

- A. 0,4
- B. 0,6
- C. 0,14
- D. 0,28

Q8

GRID-IN ALGEBRA

The function f is defined by $f(x) = mx + b$, where m and b are constants. If $f(0) = 18$ and $f(1) = 20$, what is the value of m ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

ALGEBRA

$$y = -\frac{1}{9}x$$

$$y = \frac{1}{2}x$$

The solution to the given system of equations is x, y . What is the value of x ?

- A. -9
- B. -7
- C. 0
- D. 2

Q10

ALGEBRA

Figure A and figure B are both regular polygons. The sum of the perimeter of figure A and the perimeter of figure B is 63 inches. The equation $3x + 6y = 63$ represents this situation, where x is the number of sides of figure A and y is the number of sides of figure B. Which statement is the best interpretation of 6 in this context?

- A. Each side of figure B has a length of 6 inches.
- B. The number of sides of figure B is 6.
- C. Each side of figure A has a length of 6 inches.
- D. The number of sides of figure A is 6.